

TABLE I
Commands Supported by the Camera

Command	ID number	Parameter1	Parameter2	Parameter3	Parameter4
Initial	AA01h	00h	Colour type	Raw resolution (still image only)	JPEG resolution
Get picture	AA04h	Picture type	00h	00h	00h
Snapshot	AA05h	Snapshot type	Skip frame (low byte)	Skip frame (high byte)	00h
Set package size	AA06h	00h	Package size (low byte)	Package size (high byte)	00h
Reset	AA08h	Reset type	00h	00h	XXh*
Data	AA0Ah	Data type	Length byte 0	Length byte 1	Length byte 2
Sync	AA0Dh	00h	00h	00h	00h
ACK	AA0Eh	Command ID	ACK counter	00h/package ID byte 0	00h/package ID byte 1
NAK	AA0Fh	00h	NAK counter	Error number	00h
Light	AA13h	Frequency type	00h	00h	00h

The list of main subroutines used in the IP.c source code is given in Table II.

Construction and testing

Actual-size, single-side PCB layout of the Wi-Fi embedded system is given in Fig. 8 and its component layout in Fig. 9.

Assemble the components on the PCB and connect power supply to the circuit. Before switching-on the circuit burn the hex code into the microcontroller. You need MPLAB IDE version 8.36.00.00 along with MCC18 compiler v3.84 for compiling the code and burning the hex code into the microcontroller. Take care of the following:

1. PC should be connected to either wireless modem or wireless router with modem. Note that connection between the board and router should be wireless, but the connection between PC and the router should be non-wireless. Hence PC should be connected to Ethernet port of the router.

2. The modem or router should be configured as follows:

SSID=kathir, and security=none. If you want to change SSID name, change it in `wf_config.h` file and compile and program the microcontroller.

3. Camera should be connected to the board through RS232 cable.

4. When you switch-on the circuit, you will see default IP address as 169.254.1.1. Then it will switch over to a new IP address, say, 192.168.43.51, depending on your network. You can see this address on your LCD screen also. Since DHCP client is enabled, you get the correct IP address in LCD display.

5. On typing the 192.168.43.51 IP address in address bar of your web browser, a screen will appear with 'ELECTRONICS FOR YOU' on top as shown in Fig. 10.

TABLE II
List of Main Subroutines Used in IP.c Source Code

Subroutine name	Synopsis
Main(void)	After SPI initialisation, continuously poll for incoming messages (by calling stack init()).
MACInit(void)	Initialise MRF24WB0MA registers after power up
In/output(...)	Read/write word of data to/from MRF24WB0MA registers via SPI bus interface
Stack init()	Check MRF24WB0MA for any new incoming Ethernet messages. If there are any, process them
Reply_tcp(void)	Reply to a requested TCP-IP packet. Web requests are served with this packet.
Decode_tcp(void)	Decode the type of the incoming TCP packet. This can be one of the various types to open/maintain/close a socket connection to the MRF24WB0MA based Wi-Fi webcam
Ne_send(...)	Trigger send of the already prepared Ethernet packet in the MRF24WB0MA SRAM buffer
Write_mac/ip_hdr(...)	Fill in MAC/IP header portion of new outgoing packet in the MRF24WB0MA trans buffer
Write_ip/tcp_check(...)	Read back contents of outgoing packet and append checksum in appropriate spot
Flush_print(void)	Data is written into the MRF24WB0MA one word at a time, but arrives one byte at a time. This does cleanup to flush any remaining byte
Ne_prints(...)	Write one byte of data into Ethernet packet, taking care of checksums, byte counts, etc
Finish_tcp(...)	After writing TCP packet, cleanup and send it

```
send(C328R_COMMAND_SYNC);
send(0x00);
send(0x00);
send(0x00);
send(0x00);
DelayMs(10);
}
```

The camera uses ten commands as given in Table I. In the IP.c file you can find various routines for communication with camera including `setup()`, `snap()` and `getpic()`. Camera `setup()` routine is used for configuring camera for size of the image and light

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

frequency etc. `Snap()` routine is used to take pictures and `getpic()` routine is used to retrieve pictures. `Get()` routine retrieves image data in packets and sends the packets through ENC, packet by packet, until all image data are sent. To display in HTML page we use `ne_print()` routine.